

Carson McNeill

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EDUCATION

Georgia Institute of Technology

B.S. in Computer Science | Intelligence & Information/Inter networks

Atlanta, GA

Aug 2023 - May 2027 (Expected)

- **GPA:** 4.0/4.0 | **Honors:** College of Computing Faculty Honors, Dean's List (Fall 2023 - Spring 2026)
- **Coursework:** Data Structures & Algorithms, Algorithm Design, Computer Networking, Machine Learning, AI, Databases

PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS)

Software Development Engineer Intern

Herndon, VA

Sep 2026 - Present

- Own all enhancements to an existing customer-scheduled tunnel maintenance feature for AWS Site-to-Site VPN, writing design and CX docs
- Replaced a hardcoded gap between a VPN connection's two tunnel replacements with a configurable setting in Scala, cutting downtime
- Reworked the scheduling logic to push back only the later tunnel and skip ineligible tunnels, eliminating unnecessary database checks
- Extracted notification logic into a reusable standalone Scala package and resolved 3 medium-severity security findings and a production bug

Deloitte

GPS Summer Scholar - Software Engineering

Portland, OR

Jun 2026 - Jul 2026

- Built a Power BI dashboard on an Azure Databricks medallion lakehouse, tracking individuals across 5 Oregon Health Authority databases
- Led requirements meetings with OHA stakeholders, turning their feedback into 70+ pages of technical docs and example SQL for analysts
- Built an AI documentation agent fed by Python scripts, hooks, and Markdown context, keeping dashboard docs in sync with the live report

ViTAL Lab

Undergraduate Researcher

Atlanta, GA

Aug 2025 - Present

- Enhanced a YOLOv8, OSNet ReID, and ViTPose pipeline for 41 hours of autism research video, raising F1 re-ID score by 40%
- Hand-labeled 1,000+ ground-truth images and presented validation metrics to the PI and partners, driving model and preprocessing upgrades

AgroNav Research Replication

Undergraduate Researcher

Metz, France

Jan 2025 - Apr 2025

- Replicated the *AgroNav* crop navigation pipeline, training a ResNeSt segmentation model that matched the paper's mIoU
- Trained a Deep Hough Transform line detector for crop-row centerlines with flip augmentation, nearly matching the paper's EA-score

PROJECTS

ByteFight AI Competition | C++, Python, nanobind

Mar 2026 - Apr 2026

- Peaked at 8th of ~60 teams (~2,000 Elo) in a month-long Georgia Tech bot ladder using MCTS, alpha-beta pruning, and A* search
- Moved performance-critical search into C++ with nanobind bindings and tuned evaluation weights with SPSA instead of hand-set values

GT iOS Club: SyncHub, SkillSwap

Senior Developer | Swift, Firebase, Core Spotlight

Atlanta, GA

Aug 2025 - Apr 2026

- Led a 6-developer SyncHub sub-team, assigning work, reviewing PRs, and teaching MVVM, while building Core Spotlight semantic search
- Engineered streak and badge gamification for SkillSwap in SwiftUI with MVVM architecture, shipping it at Demo Day with a team of 20+

Custom Chess Engine | Java, Python, Next.js, FastAPI

Dec 2025 - Feb 2026

- Built a Java chess engine with alpha-beta minimax, iterative deepening, and PeSTO evaluation, rated ~2,100 Elo by a FIDE Master
- Cut per-move search time 10x+ across 20+ engine versions with transposition tables, killer and null-move heuristics, and quiescence search
- Added MVV-LVA capture ordering, incremental PeSTO updates, preallocated buffers to avoid GC pauses, and clock-based time control
- Ported the engine from Python to Java for speed and built a self-play harness that pits each version against its predecessor in timed games
- Built a Next.js and FastAPI web app to play any of the 20+ Python and Java engine versions in the browser, showing search depth

HackGT 12 Winner: InfraBlocks | AWS, React, Terraform

Sep 2025

- Won Best Developer Tool out of 1,500+ participants with a drag-and-drop React Flow editor for designing AWS microservice architectures
- Built Terraform generation that turns canvas diagrams into deployable AWS microservices, replacing hand-written infrastructure code

HackGT 11: SpecTackles | Lens Studio, JavaScript, Firebase, Raspberry Pi

Sep 2024

- Placed 2nd in Snap's AR track at a 900+ hacker event by building Lens Studio marker tracking and AR buttons that toggle physical lights

TECHNICAL SKILLS

Languages

Python (NumPy, pandas, scikit-learn), Java, Scala, C/C++, TypeScript, JavaScript, Swift, SQL

Cloud & Data

AWS (EC2, EventBridge), Azure, Databricks, Power BI, Terraform, Docker, Firebase

Frameworks & Tools

React, Next.js, FastAPI, SwiftUI, PyTorch, OpenCV, YOLOv8, nanobind, Git, Linux, Slurm